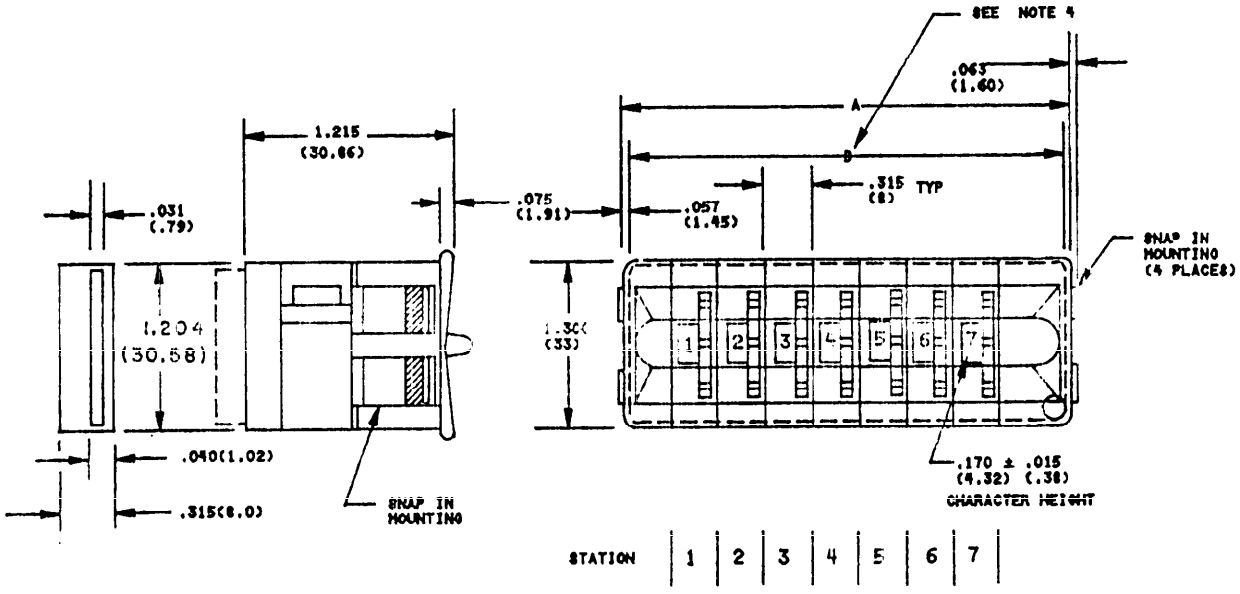
MILITARY SPECIFICATION SHEET

SWITCHES, CODE INDICATING WHEEL (PRINTED CIRCUIT), THUMBWHEEL SWITCH ASSEMBLY, INTEGRAL BEZEL, NONSEALED, NONILLUMINATED, SNAP IN MOUNTING, STYLE SRPC28

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22710.



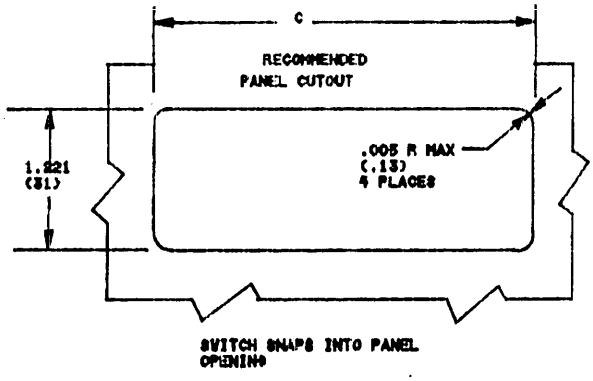


FIGURE 1. Style SRPC28 switch and panel cutout.

FSC 5930

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Number of Modules	Dimensions with RFI Shield		
	A	B	C
1	.751 (19.08)	.635 (16.13)	.657 (16.70)
2	1.068 (27.13)	.952 (24.18)	.972 (24.69)
3	1.385 (35.18)	1.269 (32.23)	1.291 (32.79)
4	1.702 (43.24)	1.586 (40.28)	1.607 (40.82)
5	2.019 (51.28)	1.903 (48.34)	1.925 (48.90)
6	2.336 (59.33)	2.220 (56.39)	2.240 (56.90)
7	2.653 (67.39)	2.537 (64.44)	2.519 (63.98)
8	2.970 (75.44)	2.854 (72.49)	2.874 (73.00)
9	3.287 (83.49)	3.171 (80.54)	3.193 (81.10)
10	3.604 (91.54)	3.488 (88.60)	3.507 (89.08)

FIGURE 1. Style SRPC 28 switch and panel cutout - Continued.

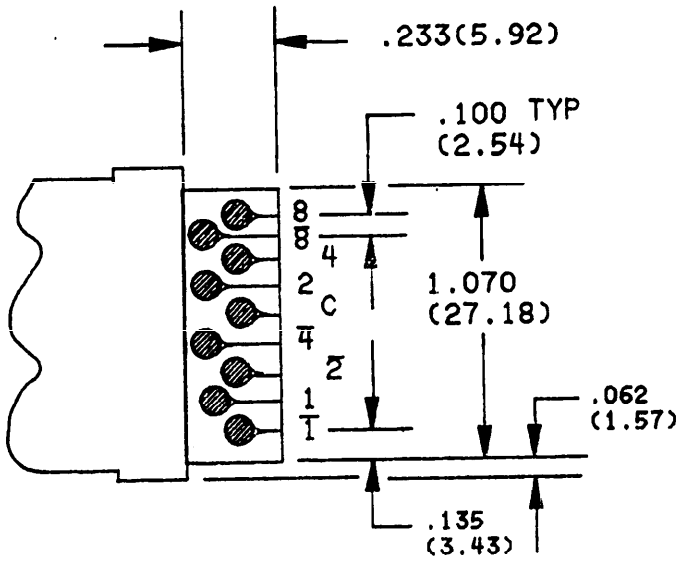
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Number of Modules	Dimensions without RFI shield		
	A ± .020 (.51)	B (SEE NOTE 4)	C
1	.748 (19.00)	.630 (16.0)	.654 (16.61)
2	1.063 (27.0)	.945 (24.0)	.969 (24.61)
3	1.378 (35.0)	1.260 (32.0)	1.283 (32.59)
4	1.693 (43.0)	1.575 (40.01)	1.598 (40.59)
5	2.008 (51.0)	1.890 (48.01)	1.913 (48.59)
6	2.323 (59.0)	2.205 (56.01)	2.228 (56.59)
7	2.638 (67.01)	2.520 (64.01)	2.543 (64.59)
8	2.953 (75.01)	2.835 (72.01)	2.858 (72.59)
9	3.268 (83.01)	3.150 (80.01)	3.173 (80.59)
10	3.583 (91.01)	3.465 (88.01)	3.488 (88.60)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
4. Dimension B refers to outside horizontal dimension of assembly behind the bezel.
5. Unless otherwise specified, tolerance is ±0.010 (0.25 mm) for three place decimals and ±0.030 (0.75 mm) for two place decimals.

FIGURE 1. Style SRPC28 switch and panel cutout - Continued

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BOARD PLATED ON BOTH SIDES
WITH PLATED THRU HOLES FOR
SOLDERED WIRE TERMINATION

HOLES=0.062 (1.57) DIA

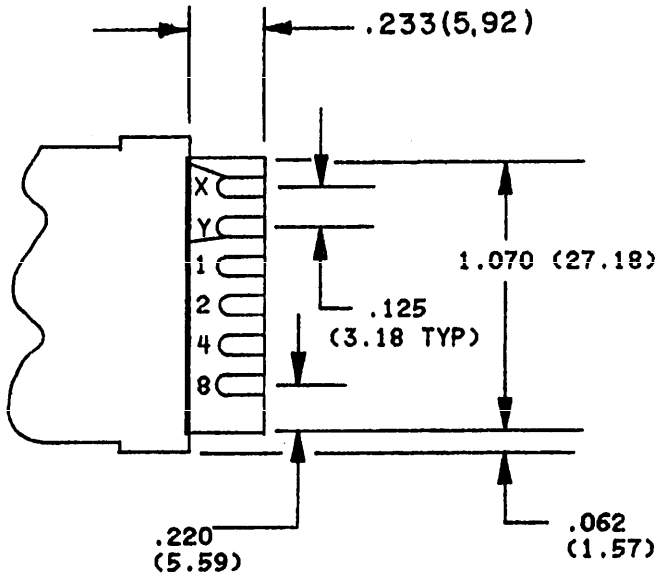
MODULE A AND H

TRUTH TABLE
(CIRCUIT CONFIGURATION 1)
36° INDEXING

DIAL MARKING	COMMON C CONN. TO							
	1	2	4	8	$\overline{1}$	$\overline{2}$	$\overline{4}$	$\overline{8}$
0					●	●	●	●
1	●					●	●	●
2		●			●		●	●
3	●	●					●	●
4			●		●	●		●
5	●		●			●		●
6		●	●		●			●
7	●	●	●					●
8				●	●	●	●	
9	●			●		●	●	

FIGURE 2. Module and circuit configuration.

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BOARD PLATED ON BOTH SIDES
WITH PLATED THRU HOLES FOR
SOLDERED WIRE TERMINATION

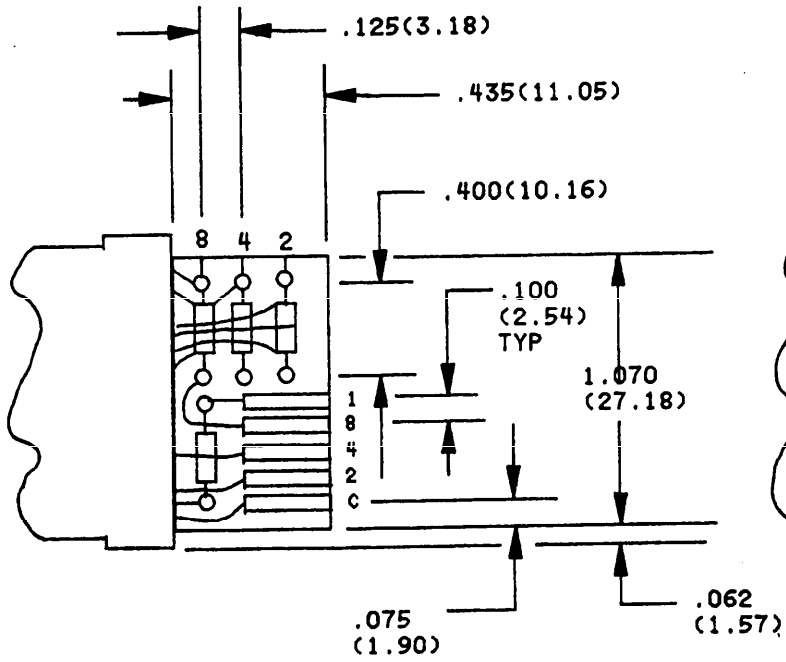
HOLES=0.062 (1.57) DIA

MODULE B AND J

TRUTH TABLE (CIRCUIT CONFIGURATION 2) 36° INDEXING						
DIAL MARKING	COMMON X(●)& Y(O) CONN. TO					
	1	2	4	8		
0	O	O	O	O		
1	●	O	O	O		
2	O	●	O	O		
3	●	●	O	O		
4	O	O	●	O		
5	●	O	●	O		
6	O	●	●	O		
7	●	●	●	O		
8	O	O	O	●		
9	●	O	O	●		

FIGURE 2. Module and circuit configuration - Continued.

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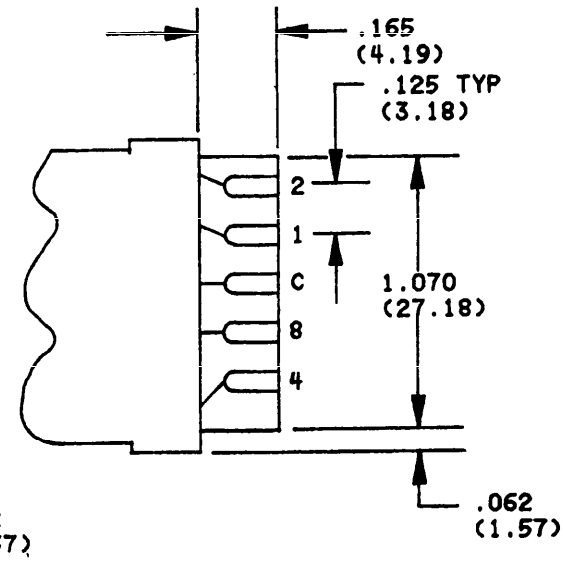


BOARD PLATED ON ONE SIDE FOR SOLDERED WIRE TERMINATION

HOLES:
COMPONENT: 0.046(1.17)DIA
WIRING : 0.046(1.17)DIA
(TWO HOLES PER TERMINATION)

EXTENDED BOARD FOR COMPONENT MOUNTING

MODULE C AND K



BOARDS PLATED ON ONE SIDE FOR SOLDERED WIRE TERMINATION

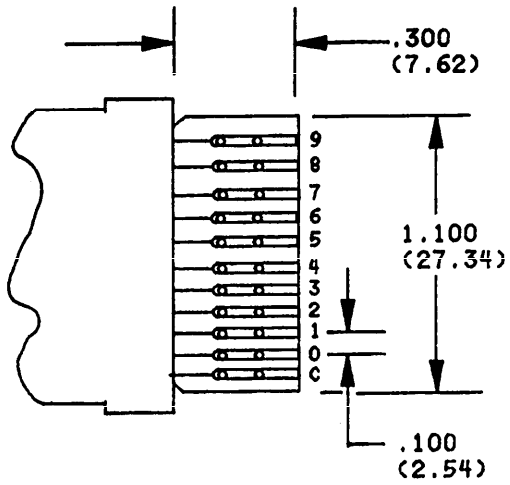
HOLES, 0.062 (1.57) DIA

MODULE D AND L

TRUTH TABLE (CIRCUIT CONFIGURATION 3) 36° INDEXING				
DIAL MARKING	COMMON C CONN. TO			
	1	2	4	8
0				
1	●			
2		●		
3	●	●		
4			●	
5	●		●	
6		●	●	
7	●	●	●	
8				●
9	●			●

FIGURE 2. Module and circuit configuration - Continued.

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BOARDS PLATED ON BOTH SIDES WITH PLATED THRU HOLES FOR SOLDERED WIRE TERMINATION

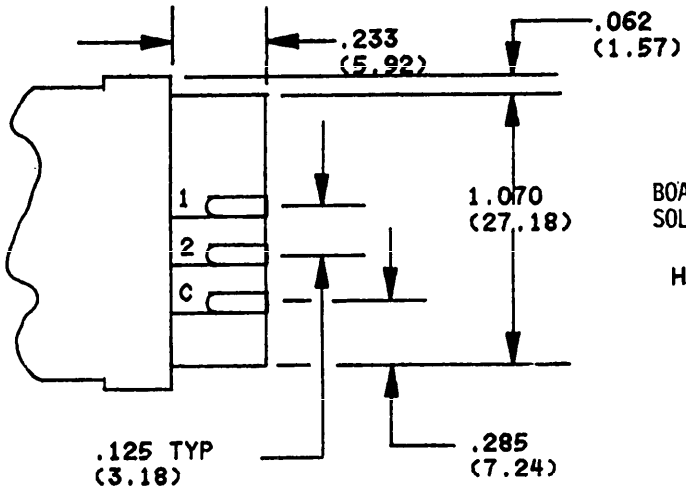
HOLES: 0.046 (1.17)DIA

MODULE E AND M

TRUTH TABLE (CIRCUIT CONFIGURATION 4) 36° INDEXING	
DIAL MARKING	COMMON C CONN. TO
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9

FIGURE 2. Module and circuit configuration - Continued.

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BOARDS PLATED ON ONE SIDE FOR SOLDERED WIRE TERMINATION

HOLES: 0.062(1.57) DIA

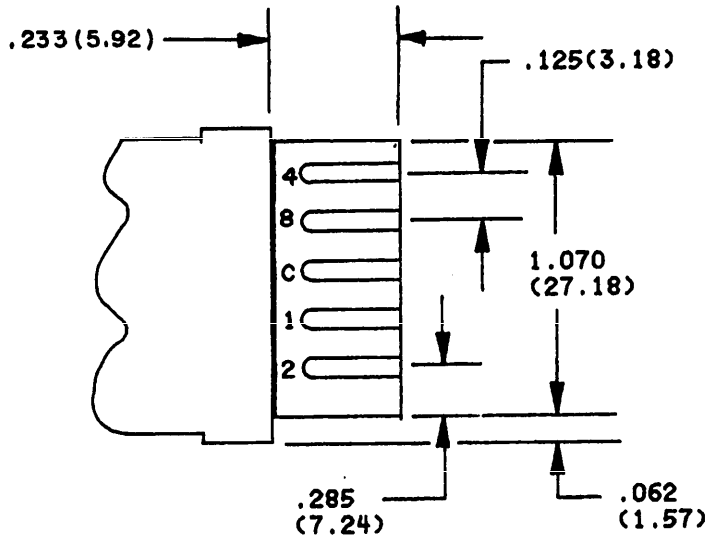
MODULE F AND N

TRUTH TABLE
(CIRCUIT CONFIGURATION 5)
36°C INDEXING

DIAL MARKING	COMMON C CONN. TO	
	1	2
0	●	
1		●
0	●	
1		●
0	●	
1		●
0	●	
1		●
0	●	
1		●

FIGURE 2. Module and circuit configuration - Continued.

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BOARD PLATED ON BOTH SIDES
WITH PLATED THRU HOLES FOR
SOLDERED WIRE TERMINATION

HOLES=0.062 (1.57) DIA

MODULE G AND P

TRUTH TABLE (CIRCUIT CONFIGURATION 6) 36° INDEXING				
DIAL MARKING	COMMON C CONN. TO			
	1	2	4	8
0		●	●	
1	●	●	●	
2				●
3	●			●
4		●		●
5	●	●		●
6			●	●
7	●		●	●
8		●	●	●
9	●	●	●	●

FIGURE 2. Module and circuit configuration - Continued.

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TABLE I. Module description. 1/ 2/

Circuit configuration	Module designation	
	RFI Unshielded	RFI Shielded
1	A	H
2	B	J
3	C, D	K, L
4	E	M
5	F	N
6	G	P

1/ All dials have black opaque background with white opaque numerals.

2/ Switch modules are not procurable by themselves, but are part of the switch assembly (see table II).

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TABLE II. Switch assembly.

Part No. M22710/28-	Switch station module number									
	1	2	3	4	5	6	7	8	9	10
001	A									
002	A	A								
003	A	A	A							
004	A	A	A	A						
005	A	A	A	A	A					
006	A	A	A	A	A	A				
007	A	A	A	A	A	A	A			
008	A	A	A	A	A	A	A	A		
009	A	A	A	A	A	A	A	A	A	
010	A	A	A	A	A	A	A	A	A	A
011	B									
012	B	B								
013	B	B	B							
014	B	B	B	B						
015	B	B	B	B	B					
016	B	B	B	B	B	B				
017	B	B	B	B	B	B	B			
018	B	B	B	B	B	B	B	B		
019	B	B	B	B	B	B	B	B	B	
020	B	B	B	B	B	B	B	B	B	B
021	C									
022	C	C								
023	C	C	C							
024	C	C	C	C						
025	C	C	C	C	C					
026	C	C	C	C	C	C				
027	C	C	C	C	C	C	C			
028	C	C	C	C	C	C	C	C		
029	C	C	C	C	C	C	C	C	C	
030	C	C	C	C	C	C	C	C	C	C
031	D									
032	D	D								
033	D	D	D							
034	D	D	D	D						
035	D	D	D	D	D					
036	D	D	D	D	D	D				
037	D	D	D	D	D	D	D			
038	D	D	D	D	D	D	D	D		
039	D	D	D	D	D	D	D	D	D	
040	D	D	D	D	D	D	D	D	D	D
041	E									
042	E	E								
043	E	E	E							
044	E	E	E	E						
045	E	E	E	E	E					
046	E	E	E	E	E	E				
047	E	E	E	E	E	E	E			
048	E	E	E	E	E	E	E	E		
049	E	E	E	E	E	E	E	E	E	
050	E	E	E	E	E	E	E	E	E	E

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TABLE II. Switch assembly - Continued.

Part No. M22710/28-	Switch station module number									
	1	2	3	4	5	6	7	8	9	10
051	F									
052	F	F								
053	F	F	F							
054	F	F	F	F						
055	F	F	F	F	F					
056	F	F	F	F	F	F				
057	F	F	F	F	F	F	F			
058	F	F	F	F	F	F	F	F		
059	F	F	F	F	F	F	F	F	F	
060	F	F	F	F	F	F	F	F	F	F
061	G									
062	G	G								
063	G	G	G							
064	G	G	G	G						
065	G	G	G	G	G					
066	G	G	G	G	G	G				
067	G	G	G	G	G	G	G			
068	G	G	G	G	G	G	G	G		
069	G	G	G	G	G	G	G	G	G	
070	G	G	G	G	G	G	G	G	G	G
071	H									
072	H	H								
073	H	H	H							
074	H	H	H	H						
075	H	H	H	H	H					
076	H	H	H	H	H	H				
077	H	H	H	H	H	H	H			
078	H	H	H	H	H	H	H	H		
079	H	H	H	H	H	H	H	H	H	
080	H	H	H	H	H	H	H	H	H	H
081	J									
082	J	J								
083	J	J	J							
084	J	J	J	J						
085	J	J	J	J	J					
086	J	J	J	J	J	J				
087	J	J	J	J	J	J	J			
088	J	J	J	J	J	J	J	J		
089	J	J	J	J	J	J	J	J	J	
090	J	J	J	J	J	J	J	J	J	J
091	K									
092	K	K								
093	K	K	K							
094	K	K	K	K						
095	K	K	K	K	K					
096	K	K	K	K	K	K				
097	K	K	K	K	K	K	K			
098	K	K	K	K	K	K	K	K		
099	K	K	K	K	K	K	K	K	K	
100	K	K	K	K	K	K	K	K	K	K

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TABLE II. Switch assembly - Continued.

Part No. M22710/28-	Switch station module number									
	1	2	3	4	5	6	7	8	9	10
101	L									
102	L	L								
103	L	L	L							
104	L	L	L	L						
105	L	L	L	L	L					
106	L	L	L	L	L	L				
107	L	L	L	L	L	L	L			
108	L	L	L	L	L	L	L	L		
109	L	L	L	L	L	L	L	L	L	
110	L	L	L	L	L	L	L	L	L	L
111	M									
112	M	M								
113	M	M	M							
114	M	M	M	M						
115	M	M	M	M	M					
116	M	M	M	M	M	M				
117	M	M	M	M	M	M	M			
118	M	M	M	M	M	M	M	M		
119	M	M	M	M	M	M	M	M	M	
120	M	M	M	M	M	M	M	M	M	M
121	N									
122	N	N								
123	N	N	N							
124	N	N	N	N						
125	N	N	N	N	N					
126	N	N	N	N	N	N				
127	N	N	N	N	N	N	N			
128	N	N	N	N	N	N	N	N		
129	N	N	N	N	N	N	N	N	N	
130	N	N	N	N	N	N	N	N	N	N
131	P									
132	P	P								
133	P	P	P							
134	P	P	P	P						
135	P	P	P	P	P					
136	P	P	P	P	P	P				
137	P	P	P	P	P	P	P			
138	P	P	P	P	P	P	P	P		
139	P	P	P	P	P	P	P	P	P	
140	P	P	P	P	P	P	P	P	P	P

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REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Construction: D.

Indexing: 36 degrees. See truth tables in figure 2.

Temperature-life characteristic: B.

Vibration grade: 3 (15g. 10-2,000 Hz).

Shock type: M (100g), Method I of MIL-S-22710 is applicable.

Altitude rating: 0 (under 10,000 feet).

Terminal design: See figure 2.

Acceleration: Applicable.

Stop Strength: 5 pounds minimum.

Moisture resistance: Method II of MIL-S-22710.

Operating force (tangential): 2 ounces minimum, 10 ounces maximum.

Explosion: Applicable

EMI/RFI Shielding: Applicable, where specified.

Life (rotational): Method 206, test condition G, MIL-STD-202 (see table III).

TABLE III. Electrical test load.

Test or condition	Resistive	
At atmospheric pressure	.050 Ampere	28 V dc
Life (low level)	10 mA	30 mV dc

Military part number: See tables I and II.

Material (body and dial): Thermoplastic per MIL-P-24519, type GPT-20F or equivalent.

CAUTION: Isopropyl alcohol recommended for cleaning and solder-flux removal. Avoid the use of acids, ammonia, chlorinated or aromatic hydrocarbons, and oxygenated solvents.

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Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, SM
Navy - AS
Air Force - 17
DLA - ES

User activities:
Army - ME
Navy - MC, OS
Air Force - 11

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5930-1176-1)

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